

Photodynamic therapy by *in situ* nonlinear photon conversion

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In photodynamic therapy, light is absorbed by a therapy agent (photosensitizer) to generate reactive oxygen, which then locally kills diseased cells. Here, we report a new form of photodynamic therapy in which nonlinear optical interactions of near-infrared laser radiation with a biological medium *in situ* produce light that falls within the absorption band of the photosensitizer. The use of near-infrared radiation, followed by upconversion to visible or ultraviolet light, provides deep tissue penetration, thus overcoming a major hurdle in treatment. By modelling and experiment, we demonstrate activation of a known photosensitizer, chlorin e6, by *in situ* nonlinear optical upconversion of near-infrared laser radiation using second-harmonic generation in collagen and four-wave mixing, including coherent anti-Stokes Raman scattering, produced by cellular biomolecules. The introduction of coherent anti-Stokes Raman scattering/four-wave mixing to photodynamic therapy *in vitro* increases the efficiency by a factor of two compared to two-photon photodynamic therapy alone, while second-harmonic generation provides a fivefold increase.

Photodynamic therapy (PDT) generally involves energy transfer from a photoexcited sensitizer molecule (PDT drug or agent) to generate highly cytotoxic singlet oxygen ($^1\text{O}_2$) to destroy a tumour^{1–6}. The scope of PDT can be significantly expanded if near-infrared (NIR) photosensitization is used, because it utilizes the biological window of maximum optical transparency and provides deep tissue penetration, enabling the treatment of remote or thick tumours⁷. However, the design and synthesis of photosensitizers that absorb light in the NIR are restricted since (i) stable NIR absorbing photosensitizers are notoriously difficult to synthesize⁷ and (ii) the triplet electronic energy level of the photosensitizer must be higher than the singlet energy level of the oxygen molecule⁸. One approach to overcoming these limitations is to make use of NIR two-photon excitation of photosensitizers^{9–13}. Because photosensitizers typically have low two-photon cross-sections, modification of their structure to produce stronger two-photon absorption, or coupling them with other strongly multiphoton absorbing species, is necessary for two-photon PDT to be successful^{14–19}. However, these approaches may affect the pharmacokinetics of the parent photosensitizer.

In this Article, we propose a radically different approach—the *in situ* generation of light by nonlinear optical interaction of incident NIR laser radiation with a natural biological medium at the wavelength that falls within the intense one-photon absorption band of the PDT agent. We demonstrate efficient PDT drug activation with NIR laser radiation and its nonlinear optical *in situ* upconversion by (1) second-harmonic generation (SHG), a second-order nonlinear optical process occurring in collagen, which is abundant in tumours^{20,21}, and (2) four-wave mixing (FWM), including coherent anti-Stokes Raman scattering (CARS), a third-order nonlinear optical process produced by biological constituents such as proteins, lipids, nucleic acids and aquatic biological environments²². CARS microscopy reveals excessive accumulation of lipids into tumorigenic lesions²³, probably caused by upregulation of lipogenic enzymes, uptake of lipoproteins through the low-density-lipoprotein receptors and/or the invasion of lipid-loaded

macrophages²⁴. These high contents of lipids will promote the susceptibility of tumours to CARS-enhanced PDT, when excitation frequencies are tuned to the vibration resonances of lipid biomolecules. Furthermore, *in situ* generation of CARS by cellular membranes^{25,26} can cause additional damage to a diverse range of subcellular organelles, where singlet oxygen enables highly localized PDT.

Propagation of NIR radiation through a biomedium and generation of photoexcitation light *in situ* within the malignancy is accompanied by minimum optical losses (due to scattering and absorption) in reaching the photosensitizer, permitting deeper treatment than with conventional PDT. *In situ* generation of spectrally tunable light by the biological medium allows for NIR excitation of a broad range of currently available photosensitizers, without affecting their structure and properties. Furthermore, *in situ* conversion of incident NIR radiation occurs in a highly localized area, increasing the spatial resolution and specificity of the PDT, which can be valuable for ophthalmologic and neurological applications.

Here, we present a theoretical assessment of the efficacy of the *in situ* generation of nonlinear optical signals for PDT initiation, as well as experimental verification of the proposed approach; these are in good agreement and validate our concept. We identify and discuss the optimization of various parameters for enhancing the proposed nonlinear PDT.

A commercial photosensitizer, chlorin e6 (ref. 27), with absorption bands peaking at ~400 nm and ~665 nm, was used to demonstrate the proposed PDT. Three different nonlinear optical upconversion mechanisms for PDT initiation were studied comparatively: (1) direct two-photon absorption (TPA) in the photosensitizer; (2) CARS and associated FWM produced by the natural intracellular macromolecules (proteins and lipids), with subsequent one-photon excitation of the photosensitizer; and (3) SHG in collagen and subsequent one-photon excitation of the photosensitizer. The nonlinear optical interactions utilized for optical signal generation are presented in Fig. 1.

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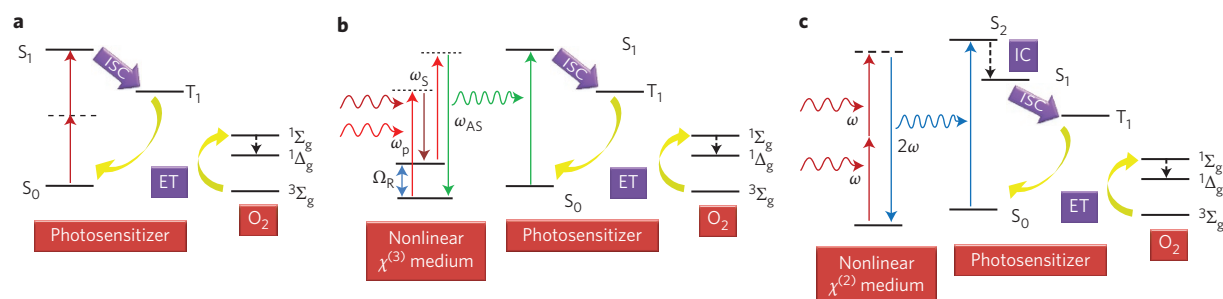


Figure 1 | Nonlinear optical mechanisms for PDT. **a–c**, Excitation with TPA (**a**), CARS/FWM (**b**) and SHG (**c**). ISC, intersystem crossing; IC, internal conversion; ET, energy transfer.

Theoretical consideration

The one-photon PDT efficiency can be modelled by using the excitation efficiency of the photosensitizer, which, in turn, equals the relative number of photo-excited molecules per unit volume. The one-photon excitation efficiency is given by

$$\zeta = \frac{\sigma_1 I_{\text{loc}}}{\Gamma \hbar \omega} \quad (1)$$

where σ_1 is the linear absorption cross-section of the photosensitizer²⁸, Γ is the lifetime broadening (half-width at half-maximum) of the excited singlet state, $\hbar$ is the reduced Planck constant, ω is the angular frequency, and I_{loc} is the local irradiance of light being absorbed by the photosensitizer molecules. I_{loc} is a function of many variables characterizing a specific nonlinear process generating light for linear absorption (Fig. 1). Without loss of generality, I_{loc} can be represented as

$$I_{\text{loc}} = I_0 \eta \quad (2)$$

where η is the conversion efficiency and I_0 is the incident irradiance. In the case of CARS/FWM, the corresponding conversion efficiency is given by²⁹

$$\eta_{\text{CARS/FWM}} = \left(\frac{\pi \omega_{\text{as}}}{n_{\text{as}} \epsilon_0 c^2} \right)^2 |3\chi^{(3)}|^2 I_0^2 L^2 \quad (3)$$

where ω_{as} is the anti-Stokes laser frequency, $\chi^{(3)}$ is the third-order nonlinear susceptibility of cellular contents^{30,31}, n_{as} is the refractive index at the anti-Stokes frequency, ϵ_0 is the permittivity of free space, c is the speed of light and L is the FWM interaction length (twice the Rayleigh range, z_R , for a tightly focused Gaussian beam). Our estimate of the interaction length utilizes a raw approximation assuming a uniform field within the interaction volume, perfect phase matching and the absence of any additional threshold. The interaction length thus depends only on the radius of the beam waist. The conversion efficiency is enhanced if molecular resonances such as in CARS are utilized, due to resonance enhancement of the

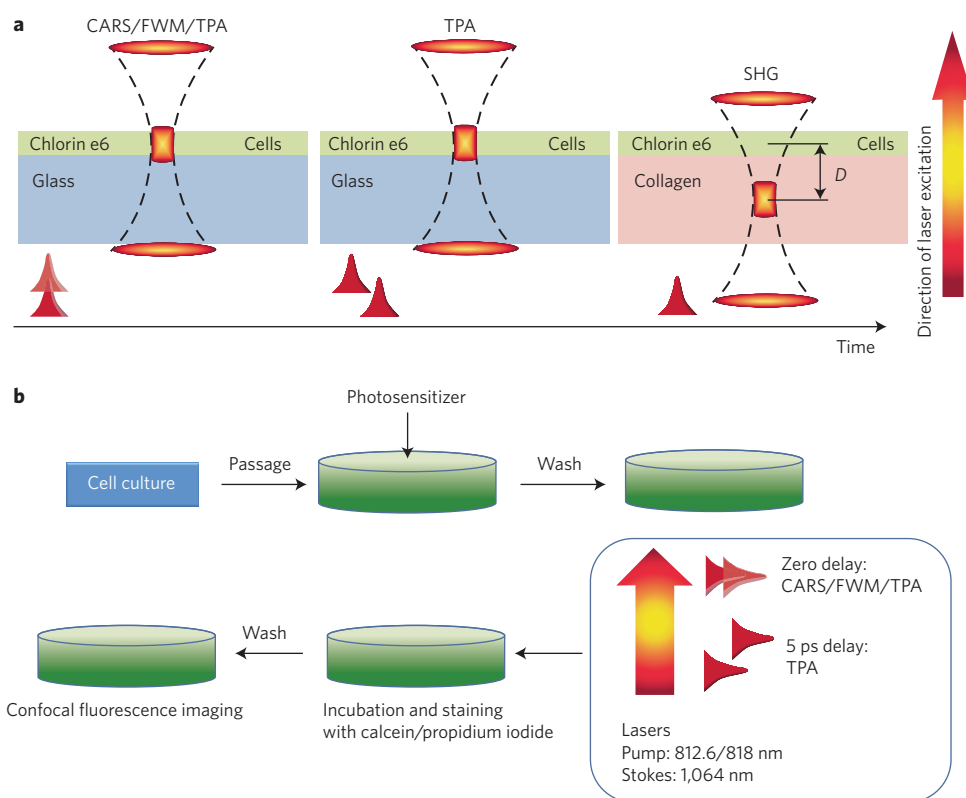


Figure 2 | Experimental design for nonlinear excitation of PDT. **a**, Nonlinear excitation diagrams. Left: laser pulses with zero delay focused on the cultured cells. Middle: delayed pulses focused in cell cultures. Right: 818 nm laser pulses focused in collagen and exciting PDT. **b**, Scheme of the experiment.

Table 1 | Chlorin e6 excitation efficiency (10^{-8}).

Gaussian beam waist radius (nm)	250	500
TPA	0.031	0.002
CARS/FWM	2.032	0.508
SHG	0.085	1.353

Laser power of 200 mW (400 mW for CARS/FWM).

electronic third-order nonlinear susceptibility, $\chi^{(3)}$ (Fig. 1b). The FWM excitation can also be accompanied by TPA (Fig. 1a). In the case of SHG (Fig. 1c), the conversion efficiency is given by

$$\eta_{\text{SHG}} = \left(\frac{2\omega_{\text{SH}}}{n^3 \varepsilon_0 c^3} \right) |\chi^{(2)}|^2 I_0 L^2 \delta \quad (4)$$

where ω_{SH} is the frequency of the second harmonic, n is the refractive index at the fundamental frequency, $\chi^{(2)}$ is the second-order nonlinear susceptibility of the medium (collagen triple helix, molecular weight of 400 kDa)³² and δ is the beam divergence parameter, equal to the squared ratio of the Gaussian beam waist w_0 to the beam radius at a distance D from the focal plane. The divergence parameter accounts for the fact that SHG takes place in the interior of the collagen layer, relatively far from the photosensitizer ($D = 12 \mu\text{m}$, Fig. 2). In contrast, CARS/FWM occurs in close vicinity to the photosensitizer, inside the cells. To compare the relative efficiencies of CARS/FWM and SHG PDT with direct two-photon PDT, we estimate the two-photon excitation efficiency³³:

$$\zeta_2 = \frac{\sigma_2 I_0^2}{2\Gamma \hbar \omega} \quad (5)$$

where σ_2 is the direct (simultaneous³⁴) TPA cross-section of the photosensitizer. Equations (1) and (5) are derived under the assumption of quasi-continuous-wave (c.w.) excitation and moderate irradiances (200 mW average power) to exclude saturation effects³³. The results are summarized in Table 1.

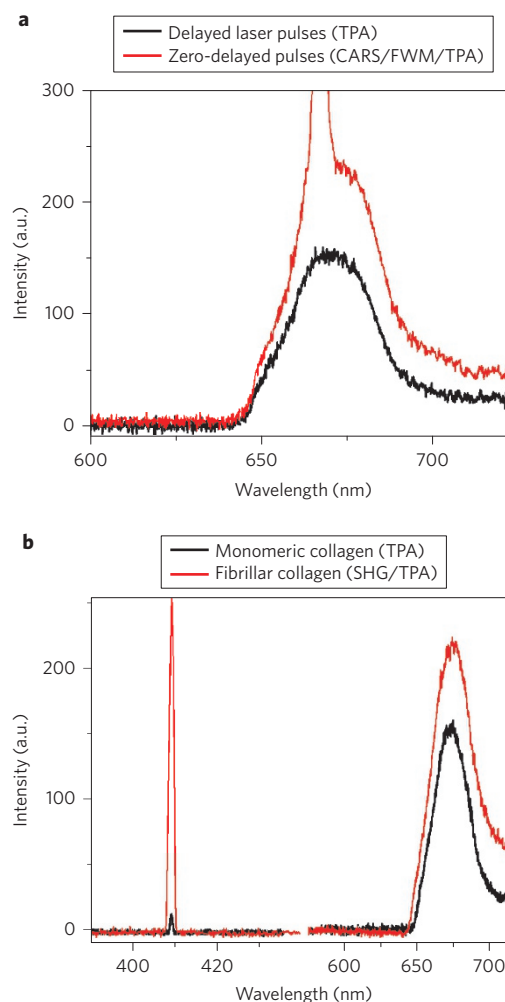
Here, we used the chlorin-e6 TPA cross-section, computed with time-dependent density functional theory at the CAM-B3LYP/6-31G* level in Dalton³⁵ in the gas phase, with the geometry optimized in vacuum at the HF/6-31 + G* level. This computed value is twice the experimental value reported at 800 nm (ref. 36).

It should be noted that in the case of PDT induced by SHG, the second excited singlet state S_2 of chlorin e6 is populated, as opposed to S_1 as in the case of CARS/FWM (Fig. 1). Because internal conversion (IC) of S_2 to S_1 is in general fast (on the order of picoseconds), this difference in initial excitation should not matter. In the case of a tightly focused beam (waist radius of 250 nm), CARS/FWM generates a more efficient photoexcitation of the photosensitizer according to our model. When the focal spot is enlarged (500 nm radius), SHG begins to dominate, although CARS/FWM is still pronounced. In both cases, the contribution of *in situ* generated light to the excitation of chlorin e6 greatly prevails over the contribution of TPA.

Experimental verification

In the present experiment, the fluorescence intensity of chlorin e6 serves as an indicator of the excitation efficiency of the photosensitizer and, to a first approximation, the efficiency of the PDT. Chlorin e6 was diluted in the culture medium and its fluorescence intensity was measured under different experimental settings.

In the first series of experiments, cell medium containing chlorin e6 was placed in a dish with the bottom covered with an $\sim 10 \mu\text{m}$ layer of lipids, and excited by focusing coinciding dual beams at 1,064 nm (Stokes wave) and 818 nm (pump wave tuned to the vibrational resonance of the lipids at $2,840 \text{ cm}^{-1}$, corresponding to CH_2 symmetric stretching) on the lipid layer (see Methods).

**Figure 3 | Contributions of CARS/FWM and SHG nonlinear optical conversions to chlorin e6 excitation.**

a, b. Fluorescence signals from buffer solution of chlorin e6 on top of a thin lipid layer, with excitation by 818 and 1,064 nm laser pulses with zero delay to achieve a resonance anti-Stokes CARS/FWM signal at 665 nm (red curve) or with laser pulses delayed by 5 ps (black curve) (**a**), and on top of a thin collagen gel layer that generates a strong SHG signal at 409 nm (red spectrum) or over monomeric collagen (black spectrum), which generates only negligible levels of SHG and produces a fluorescence signal only by TPA excitation of buffer solution (black) (**b**).

Laser pulses were synchronized in time, either with zero delay to achieve the CARS/FWM/TPA mode, or with a 5 ps time delay to disable CARS/FWM interactions in the lipid layer, while retaining the TPA contribution. The emitted fluorescence of chlorin e6 was detected with a fibre-coupled spectrometer. For the laser pulses with zero delay, the emitted fluorescence signal was higher than for the laser pulses with a 5 ps delay for the same incident beam intensity (Fig. 3a). The enhancement of chlorin e6 fluorescence with zero-delay laser pulses originates from the efficient single photon absorption of the nonlinear CARS/FWM signal produced by the lipid layer at the anti-Stokes wavelength of $\sim 665 \text{ nm}$, which corresponds to a maximum in the chlorin e6 absorption band. In a similar manner, protein vibration resonance with the CARS/FWM anti-Stokes wavelength at 657 nm could be applied to increase the chlorin e6 fluorescence and enhance the PDT efficiency.

In a second series of experiments, we studied the excitation of chlorin e6 by SHG in collagen with a laser beam at 818 nm.

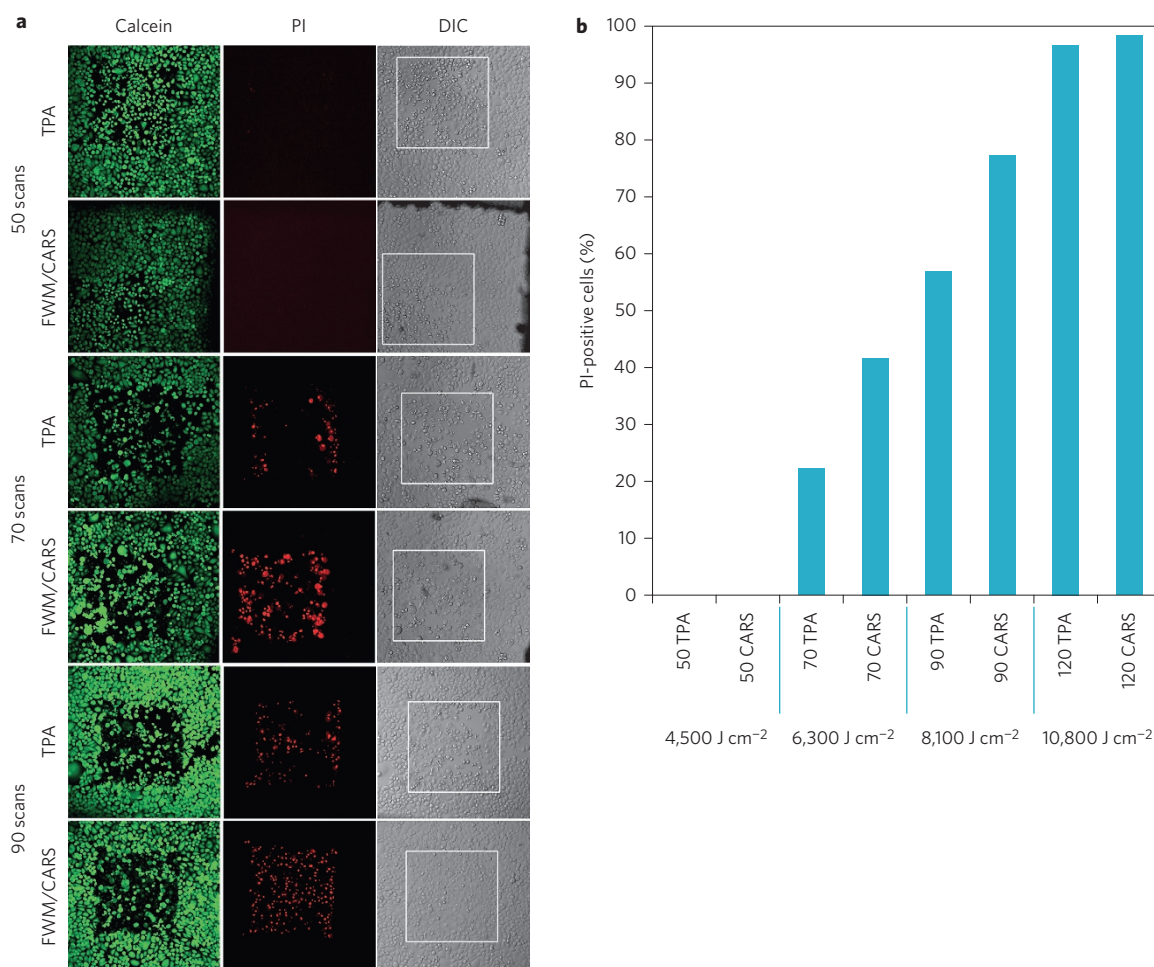


Figure 4 | Cellular phototoxicity in response to nonlinear optical PDT excitation by CARS/FWM/TPA and TPA modes for different numbers of laser scans. a, Fluorescence images of calcein/PI staining and corresponding DIC panels of chlorin-e6-treated cells irradiated by 50, 70 and 90 scans. **b**, Percentage of necrotic (PI-positive) cells at various scan numbers for CARS/FWM/TPA and TPA nonlinear optical modes, averaged over four series of experiments.

Chlorin-e6-buffered solution was placed on top of either polymerized collagen gels known to produce strong SHG on their quasi-ordered structure or on top of monomeric collagen layers that produce only negligible SHG (Fig. 3b). The SHG wavelength of 409 nm falls into the strongest Soret absorption band of chlorin e6 and contributes to excitation of the fluorophore (red fluorescence peak in Fig. 3b). In the absence of SHG, a fluorescence offset in Fig. 3b (black curve) is produced by direct TPA of chlorin e6 in the buffered solution of the monomeric collagen sample. Thus, our results confirm that nonlinear optical interactions (CARS/FWM and SHG) between the intense laser radiation and the natural biological medium enable photoexcitation of the photosensitizer.

This concept was experimentally validated by modelling the PDT treatment in live HeLa cells growing in ~90% confluent monolayer cultures. The experimental nonlinear optical set-up (Supplementary Fig. 1) was used for regulated irradiation of cell culture samples for PDT and for cell imaging.

To assess any cytotoxicity caused by the laser radiation itself, cultured cells were scanned with the focused pump and Stokes laser beams with zero pulse delay, and photodamage was evaluated by calcein AM and propidium iodide (PI) cellular viability assay (Supplementary Materials). In the absence of the photosensitizer, there were no signs of cytotoxicity caused by picosecond light pulses. The cells, grown either on glass or on a collagen layer, did not incorporate PI and showed no visible changes in the intensity

of calcein fluorescence in response to irradiation by up to 200 sequential laser scans (Supplementary Fig. 2), thereby demonstrating the low cytotoxicity from the NIR and upconverted radiation, in agreement with previous reports^{25,26,37}. We therefore concluded that the incident NIR radiation and generated *in situ* CARS/FWM and SHG light did not cause noticeable photo- and thermally induced cytotoxicity in the absence of photosensitizer. However, some cytotoxicity in the presence of the photosensitizer due to a small increase in temperature (thermally enhanced PDT) is possible³⁸.

To evaluate the contributions of CARS/FWM and SHG processes to the PDT effect *in vitro* we performed the following studies. Cells grown on the glass window of the dishes were treated with chlorin e6 and irradiated by a series of scans of dual-beam laser pulses. The pump wave was tuned to 818 nm, corresponding to the vibration resonance of lipids at $2,840 \text{ cm}^{-1}$. Samples were scanned either with pump and Stokes laser pulses with zero delay to effect CARS/FWM/TPA or with a delay to trigger only TPA. The experimental data regarding the cytotoxic responses of the treated cells are presented in Fig. 4. The fluorescence images of calcein and PI cellular staining, together with differential interference contrast (DIC) transmission light images for different irradiation doses (scan numbers), illustrate the PDT effect. Both nonlinear optical conversion mechanisms induced PDT in the irradiated area. Cellular phototoxicity increased with radiation dose. The increase

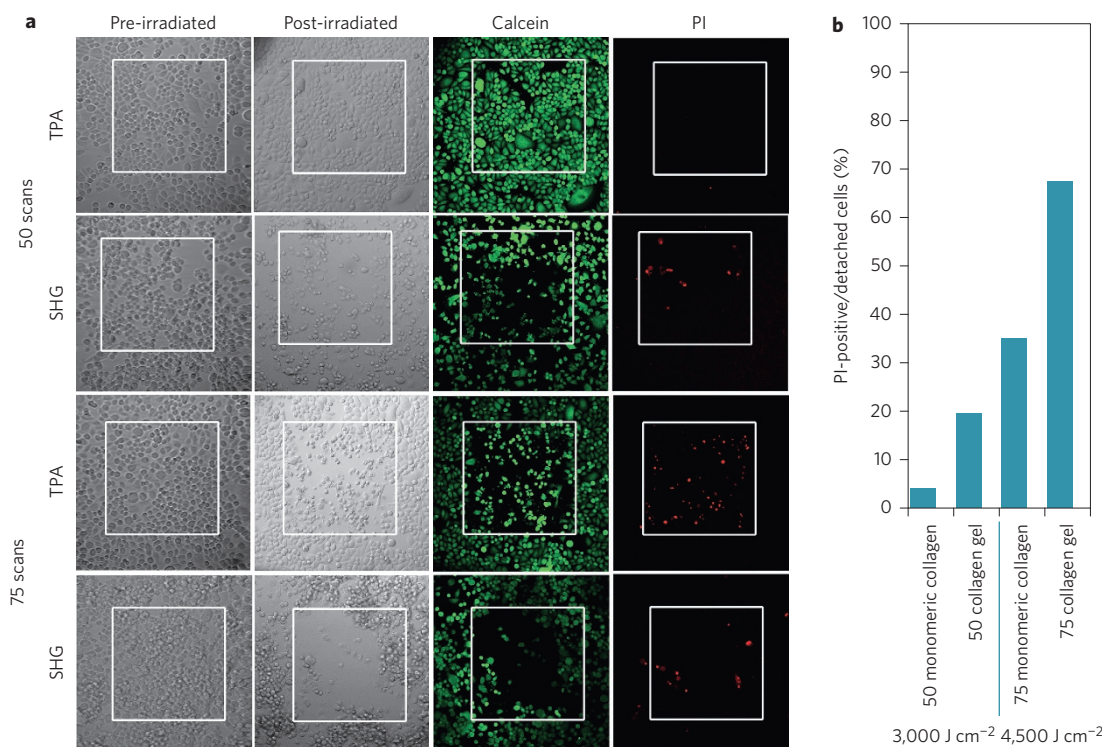


Figure 5 | Cellular phototoxicity in response to nonlinear optical PDT excitation by SHG/TPA and TPA modes for different numbers of laser scans. a, Cells grown either on monomeric collagen or polymerized collagen gel were irradiated for different scan numbers. Left to right: transmission light images of cells before PDT initiation, as well as DIC and calcein/PI fluorescence images of irradiated cells, as indicated. **b,** Percentage of necrotic (PI-positive) and detached cells at various scan numbers for SHG/TPA and TPA nonlinear optical modes, averaged over four series of experiments.

in the number of laser scans initially diminished the fluorescence intensity of the calcein signal, pointing to a decrease in cellular metabolic activities and/or to disturbance of the cellular membranes. Further increasing the irradiation dose led to the rapid development of necrosis, as indicated by the increased number of cells positively stained with PI (Fig. 4a).

We next addressed the SHG conversion mechanism. Its practical value is apparent, as the stroma of solid tumours frequently exhibits extensive deposits of collagen fibres, which are known to produce very strong SHG³⁹. SHG imaging has already been used broadly for diagnosis, prognostic analysis and monitoring of therapies in solid tumours⁴⁰. To study PDT enhancement by SHG nonlinear optical conversion in a collagen ordered structure, dishes were coated either with collagen fibre gels producing strong SHG signals (Fig. 3b), or with monomeric collagen, which generates negligible SHG. Cells growing on both types of collagen substrate were incubated with chlorin e6 and irradiated with the 818 nm beam focused in the collagen layer (Fig. 2a). Excitation with a single laser beam was used in this experiment to identify the input of SHG nonlinear optical conversion to PDT excitation, eliminating other nonlinear optical mechanisms such as resonance CARS, FWM and SHG from the 1,064 nm beam. As discussed above, the SHG signal at 409 nm (falling within the intense Soret absorption band of chlorin e6) can trigger PDT by the generation of SHG in the collagen layer and its subsequent propagation and absorption by the photosensitizer inside cells. In addition, the photosensitizer was concurrently excited by direct TPA due to the dimensions of the laser waist exceeding the thickness of the collagen layer. Also, a contribution from weak collagen autofluorescence with a broad spectrum overlapping with chlorin e6 absorption⁴¹ could play a role in PDT excitation.

The results of the phototoxicity study are shown in Fig. 5. It is worth noting that the cells growing on the collagen substrates and

treated with chlorin e6 are more easily detached in response to laser irradiation than treated cells growing directly on the uncoated glass surface. Therefore, for a quantitative analysis of phototoxicity, we counted both detached and PI-positive cells as photodamaged.

Discussion

To quantify the efficacy of the PDT treatment induced by drug excitation through non-resonant nonlinear optical interaction of incident NIR laser radiation we used cell phototoxicity assays (Figs 4 and 5).

Figure 4b plots the percentage of PI positive cells against the number of scans for the two studied nonlinear optical modes, averaged over four series of experiments. The cytotoxicity threshold of CARS/FWM/TPA phototherapy was reached at irradiation doses exceeding $4,500 \text{ J cm}^{-2}$ (50 laser scans in Fig. 4). At a radiation dose of $\sim 6,300 \text{ J cm}^{-2}$, delivered to the cells in 70 scans, the efficiency of PDT induced by the CARS/FWM/TPA mechanism surpasses twice the efficiency of phototherapy activated by TPA alone. At this irradiation dose we observed a positive PI signal in 40% of cells for the CARS/FWM/TPA excitation mode, and observed 20% necrotic cells when treated with the same number of scans in the TPA mode. Further increasing the irradiation dose (number of scans) leads to saturation of the treatment process, and the difference in the number of damaged cells for the two excitation modes decreases, although the higher efficiency of CARS/FWM/TPA is still notable for 90 scans (Fig. 4a,b). We thus conclude a significantly higher efficiency and lower threshold for PDT in the CARS/FWM/TPA mode when compared to the conventional TPA approach.

It should be noted that tuning the CARS excitation wavelengths to match the specific resonance frequencies of proteins, lipids or nucleic acids provides an elegant mechanism for selective excitation of a photosensitizer associated with different structural elements of the treated tissue. In particular, using lipid CARS resonance can be

very useful for PDT of the many types of tumorigenic lesions known to accumulate lipids^{23,24}.

In the sample with cells growing on the layers of fibrillar collagen, after 50 scans with an 818 nm laser (corresponding radiation dose of $\sim 3,000 \text{ J cm}^{-2}$) we observed a substantial decrease of calcein signal intensity. We also documented positive PI staining and detachment from the substrate in 20% of the cells, on average. In contrast, the monomeric collagen demonstrated less than 5% of necrotic or detached cells at the same irradiation dose (Fig. 5a,b). This significant difference in phototoxicity between the two experimental conditions confirms the high efficiency of SHG nonlinear optical conversion for chlorin e6 PDT action. When the irradiation dose was increased to 75 scans (corresponding to $\sim 4,500 \text{ J cm}^{-2}$), we found that $\sim 35\%$ of cells growing at the monomeric collagen either detached or displayed PI staining, showing a still lower level of photodamage compared to the $\sim 70\%$ of detached and PI-positive cells growing on the fibrillar collagen substrates. We therefore conclude that SHG by collagen fibrils contributes efficiently to the excitation of chlorin e6 for phototherapy. It is worth noting that no chlorin e6 was found associated with the collagen fibrils, as shown by the absence of its fluorescence signal. Therefore, all cytotoxic effects were caused by the photosensitizer taken up by cells. Considering the significant SHG nonlinear optical conversion demonstrated here, it is important to note that many reports on conventional TPA-induced PDT^{10,11} utilizing pico- or femtosecond Ti:sapphire lasers at excitation wavelengths in the range 750–850 nm could also have contributions from the SHG signal at $\sim 400 \text{ nm}$ generated by fibrillar collagens in malignant tissues.

In conclusion, we have presented theoretical justification and experimental demonstration of novel nonlinear optical upconversion mechanisms for PDT utilizing CARS, FWM and SHG processes in intra- or extracellular native biomolecules. Consistent with the theoretical modelling (Table 1), we find that the input of SHG, when compared to the other types of nonlinear optical conversion, appears to be more significant for PDT excitation. The experimental results show that the threshold for irreversible photodamage under SHG/TPA excitation is ~ 1.5 times lower than that for CARS/FWM/TPA excitation. With the same irradiation dose ($4,500 \text{ J cm}^{-2}$), CARS/FWM/TPA excitation was on the threshold level of phototoxicity, but in the SHG/TPA mode, nearly 70% of cells were either necrotic or detached, indicating a higher efficiency for the nonlinear SHG/TPA approach in comparison to CARS/FWM/TPA, let alone TPA. These approaches can be used as a complement to conventional two-photon PDT, accumulating treatment benefits from all nonlinear optical mechanisms. The combination of such newly proposed nonlinear optical excitation techniques with the already well-developed two-photon PDT technology will enable highly selective, high-resolution, more deeply penetrating enhanced PDT treatment with a lower radiation threshold.

Methods

Photosensitizer excitation. The optical parametric oscillator frequency was tuned to 812.6 nm or 818.0 nm to align the laser excitation to match the vibration resonance of proteins ($2,930 \text{ cm}^{-1}$) or lipids ($2,840 \text{ cm}^{-1}$)²⁵. In these conditions the anti-Stokes emission frequency $\omega_{\text{as}} = 2\omega_{\text{p}} - \omega_{\text{s}}$ corresponds to wavelengths of 657 nm or 665.4 nm, which overlap with the absorption band of photosensitizer chlorin e6 in the visible range of the spectrum (maximum at $\sim 665 \text{ nm}$ for a solution in dimethylsulphoxide). As a rule, the resonance signal of CARS from molecules of proteins or lipids is also accompanied by a non-resonance FWM signal at the same ω_{as} frequencies. The incident excitation radiation ω_{p} at 812.6 nm and 818.0 nm closely matches the corresponding wavelengths of the direct TPA band of photosensitizer chlorin e6. Therefore, appropriate TPA, together with CARS/FWM, will supply further PDT enhancement. Finally, the second-harmonic signal of the ω_{p} wave lies in the short wavelength range of the spectrum around 400 nm, and coincides well with the most intense Soret absorption peak of chlorin e6 and could also be used for more efficient excitation of PDT.

The SHG-induced excitation of chlorin e6 was tested in dishes covered with an $\sim 10 \mu\text{m}$ layer of either polymerized (fibrillar) or monomeric (Gibco) collagen.

Polymerization of the collagen gel was performed according to the manufacturer's instructions and air-dried. Excitation of chlorin e6 by CARS was tested in dishes with their bottoms covered with an $\sim 10\text{-}\mu\text{m}$ -thick lipid layer, prepared by drop-casting from a stock solution of the lipids (bovine heart extraction, Avanti) in chloroform (50 mg ml^{-1}).

To verify the efficiency of the different nonlinear optical conversion mechanisms for PDT excitation, cell cultured samples were used in different layouts, permitting the separation of different nonlinear processes. The nonlinear excitation layout diagram is shown in Fig. 2a. All experiments were made in Petri dishes with optical-glass bottom windows (MatTek). When cells were grown directly on the optical glass window, the pump and Stokes waves were synchronized by a delay line to zero delay and the beams were focused into the cells, and CARS/FWM signals were generated on the natural intracellular biomolecules of proteins or lipids. In this case, CARS/FWM conversion was also supplemented by the TPA process, directly in the photosensitizer, and a combination of CARS/FWM/TPA for drug excitation was studied (Fig. 2a, left). The introduction of a pulse delay of $\sim 5 \text{ ps}$ between pump and Stokes waves disabled CARS/FWM, and PDT excitation was produced by TPA (Fig. 2a, middle).

An additional experimental layout was used where cells were grown on top of an $\sim 10 \mu\text{m}$ layer of polymerized collagen gels (ordered collagen structures). In this case, the Stokes laser beam was shut down while the pump beam (818 nm) was focused in the collagen substrate, where nonlinear interactions occurred, producing SHG (Fig. 2a, right; inverted microscope configuration). The SHG radiation generated in the collagen forward-propagated towards the cells treated with photosensitizer to induce PDT (Fig. 2a, right).

The dose of laser radiation delivered to the sample was estimated to be $\sim 60 \text{ J cm}^{-2}$ at 812.6 nm or 818.8 nm and $\sim 30 \text{ J cm}^{-2}$ at 1,064 nm, per scan. Thus, the total dose per scan was $\sim 90 \text{ J cm}^{-2}$ for CARS/FWM and $\sim 60 \text{ J cm}^{-2}$ for SHG. These doses of incident NIR radiation in the absence of photosensitizer did not cause any significant thermally induced cytotoxicity, as shown by our control experiment (Supplementary Fig. 2).

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Author contributions

A.V.K. provided technical and conceptual support and edited the manuscript. A.P. performed experiments, analysed data and wrote the manuscript. A.N.K. designed and performed experiments, analysed data and edited the manuscript. A.B. performed theoretical modelling and wrote the theoretical analysis. T.Y.O. designed experiments, analysed the data and wrote the manuscript. J.Q. performed experiments and edited the manuscript. P.N.P. supervised the analysis and edited the manuscript.

Additional information

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Competing financial interests

The authors declare no competing financial interests.